

	$\mathcal{K}_0^{#1}$	$\mathcal{K}_0^{#1} \alpha_{\beta\chi}$				
$\mathcal{K}_0^{#1} \dagger$	$\frac{1}{4} (-\gamma_1 k^2 - 6 \binom{(2)}{\kappa_3})$	0				
$\mathcal{K}_0^{#1} \dagger^{\alpha\beta\chi}$	0	$3 k^2 \binom{(4)}{\kappa_{15}}$	$\mathcal{K}_1^{#1} \alpha_{\beta}$	$\mathcal{K}_1^{#2} \alpha_{\beta}$	$\mathcal{K}_1^{#1} \alpha$	$\mathcal{K}_1^{#2} \alpha$
	$\mathcal{K}_1^{#1} \dagger^{\alpha\beta}$	$-\frac{\gamma_2 k^2}{4}$	$\frac{\gamma_2 k^2}{2 \sqrt{2}}$	0	0	
	$\mathcal{K}_1^{#2} \dagger^{\alpha\beta}$	$\frac{\gamma_2 k^2}{2 \sqrt{2}}$	$-\frac{\gamma_2 k^2}{2}$	0	0	
	$\mathcal{K}_1^{#1} \dagger^{\alpha}$	0	0	$-k^2 \binom{(4)}{\kappa_2} - \binom{(2)}{\kappa_3}$	$\frac{\gamma_3 k^2 + 2 \binom{(2)}{\kappa_3}}{2 \sqrt{2}}$	
	$\mathcal{K}_1^{#2} \dagger^{\alpha}$	0	0	$\frac{\gamma_3 k^2 + 2 \binom{(2)}{\kappa_3}}{2 \sqrt{2}}$	$\frac{1}{2} (-\gamma_4 k^2 - \binom{(2)}{\kappa_3})$	$\mathcal{K}_2^{#1} \alpha_{\beta} \quad \mathcal{K}_2^{#1} \alpha_{\beta\chi}$
				$\mathcal{K}_2^{#1} \dagger^{\alpha\beta}$	$\frac{\gamma_5 k^2}{4}$	0
				$\mathcal{K}_2^{#1} \dagger^{\alpha\beta\chi}$	0	0

[illegible]

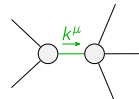
$$\begin{aligned} & \text{Abbreviations used in matrices} \\ Y_1 &= 6 \binom{(4)}{\kappa_1} - 12 \binom{(4)}{\kappa_{15}} + 6 \binom{(4)}{\kappa_2} + 4 \binom{(4)}{\kappa_3} - \binom{(4)}{\kappa_4} \&\& Y_2 = 4 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_4} \&\& \\ Y_3 &= 2 \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_7} \&\& Y_4 = 6 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_2} + \binom{(4)}{\kappa_3} + 2 \binom{(4)}{\kappa_4} + \binom{(4)}{\kappa_7} \&\& Y_5 = 12 \binom{(4)}{\kappa_{15}} - 4 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4} \&\& \\ \text{Det}(0^+) &= \frac{1}{4} (\kappa^2 (-6 \binom{(4)}{\kappa_1} + 12 \binom{(4)}{\kappa_{15}} - 6 \binom{(4)}{\kappa_2} - 4 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) - 6 \binom{(2)}{\kappa_3}) \&\& \\ \text{Det}(1^+) &= -\frac{3}{2} \kappa^2 (4 \binom{(4)}{\kappa_{15}} + \binom{(4)}{\kappa_4}) \&\& \text{Det}(2^+) = \frac{1}{2} \kappa^2 (12 \binom{(4)}{\kappa_{15}} - 4 \binom{(4)}{\kappa_3} + \binom{(4)}{\kappa_4}) \end{aligned}$$

$$\begin{aligned} & \text{Lagrangian} \\ & {}^{(2)}_3 \mathcal{K}^{\alpha\beta}{}_{\alpha} \mathcal{K}^{\chi}{}_{\beta\chi} + {}^{(4)}_1 \partial_{\beta} \mathcal{K}^{\delta}{}_{\chi\delta} \partial^{\chi} \mathcal{K}^{\alpha\beta}{}_{\alpha} + {}^{(4)}_2 \partial_{\chi} \mathcal{K}^{\delta}{}_{\beta\delta} \partial^{\chi} \mathcal{K}^{\alpha\beta}{}_{\alpha} + {}^{(4)}_3 \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}{}_{\alpha\chi} + {}^{(4)}_4 \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}{}_{\beta\chi} - \\ & 6 {}^{(4)}_{15} \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}{}_{\chi\alpha} + {}^{(4)}_5 \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}{}_{\chi\alpha} - \frac{1}{2} {}^{(4)}_4 \partial_{\beta} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}{}_{\chi\alpha} + {}^{(4)}_7 \partial^{\chi} \mathcal{K}^{\alpha\beta}{}_{\alpha} \partial_{\delta} \mathcal{K}^{\delta}{}_{\chi\beta} - \\ & 3 {}^{(4)}_{15} \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}{}_{\beta\chi} - \frac{1}{2} {}^{(4)}_4 \partial_{\alpha} \mathcal{K}^{\alpha\beta\chi} \partial_{\delta} \mathcal{K}^{\delta}{}_{\beta\chi} + {}^{(4)}_{15} \partial_{\delta} \mathcal{K}^{\alpha\beta\chi} \partial^{\delta} \mathcal{K}^{\alpha\beta\chi} - 2 {}^{(4)}_{15} \partial_{\delta} \mathcal{K}^{\alpha\beta\chi} \partial^{\delta} \mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):		$\mathcal{K}^{\alpha\beta\chi} \mathcal{T}_{\alpha\beta\chi}$
Source constraint(s)	# constraint(s)	Covariant form
$2 \mathcal{T}_{1^{+}}^{\#1\alpha\beta} + \mathcal{T}_{1^{+}}^{\#2\alpha\beta} == 0$	3	$\partial_{\chi} \mathcal{T}^{\alpha\beta\chi} + \partial_{\chi} \mathcal{T}^{\chi\alpha\beta} == \partial_{\chi} \mathcal{T}^{\beta\alpha\chi}$
$\mathcal{T}_2^{\#1\alpha\beta\chi} == 0$	5	$3 \partial_{\epsilon} \partial_{\delta} \partial^{\alpha} \partial^{\chi} \mathcal{T}^{\delta\beta\epsilon} + 3 \partial_{\epsilon} \partial^{\epsilon} \partial^{\alpha} \partial^{\chi} \mathcal{T}^{\delta\beta}_{\delta} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{T}^{\chi\delta}_{\delta} + 4 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{T}^{\chi\delta\epsilon} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\beta} \mathcal{T}^{\delta\alpha\chi} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{T}^{\beta\delta\chi} +$ $4 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\chi} \mathcal{T}^{\delta\alpha\beta} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{T}^{\delta\chi\beta} + 3 \eta^{\beta\chi} \partial_{\phi} \partial^{\phi} \partial_{\epsilon} \partial^{\alpha} \mathcal{T}^{\delta\epsilon}_{\delta} + 3 \eta^{\alpha\chi} \partial_{\phi} \partial^{\phi} \partial_{\epsilon} \partial_{\delta} \mathcal{T}^{\delta\beta\epsilon} + 3 \eta^{\beta\chi} \partial_{\phi} \partial^{\phi} \partial_{\epsilon} \partial^{\delta} \mathcal{T}^{\delta\alpha}_{\delta} ==$ $3 \partial_{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{T}^{\delta\beta\epsilon} + 3 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{T}^{\delta\beta}_{\delta} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{T}^{\beta\chi\delta} + 4 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{T}^{\chi\beta\delta} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{T}^{\delta\beta\chi} + 2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\alpha} \mathcal{T}^{\chi\alpha\beta\delta} +$ $2 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\delta} \mathcal{T}^{\beta\alpha\chi} + 4 \partial_{\epsilon} \partial^{\epsilon} \partial_{\delta} \partial^{\delta} \mathcal{T}^{\chi\alpha\beta} + 3 \eta^{\alpha\chi} \partial_{\phi} \partial^{\phi} \partial_{\epsilon} \partial^{\beta} \mathcal{T}^{\delta\epsilon}_{\delta} + 3 \eta^{\beta\chi} \partial_{\phi} \partial^{\phi} \partial_{\epsilon} \partial_{\delta} \mathcal{T}^{\delta\alpha\epsilon} + 3 \eta^{\alpha\chi} \partial_{\phi} \partial^{\phi} \partial_{\epsilon} \partial^{\delta} \mathcal{T}^{\delta\beta}_{\delta}$

Total # constraint(s):	8		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{192 \binom{(4)}{k_{15}}^2 - 48 \binom{(4)}{k_{15}} \binom{(4)}{k_3+4} \binom{(4)}{k_3}^2 + 72 \binom{(4)}{k_{15}} \binom{(4)}{k_4-8} \binom{(4)}{k_3} \binom{(4)}{k_4} + 7 \binom{(4)}{k_4}^2}{\binom{(4)}{k_{15}+k_4} \binom{(4)}{12} \binom{(4)}{k_{15}-4} \binom{(4)}{k_3+k_4} \binom{(4)}{6} \binom{(4)}{k_{15}+k_3} + 2 \binom{(4)}{k_4}}$

	1	0	$ \begin{aligned} & (1728 \overset{(4)}{K}_{15} \overset{(2)}{K}_3 + 576 \overset{(4)}{K}_{15} \overset{(2)}{K}_3 \overset{(2)}{K}_3 + 48 \overset{(4)}{K}_{15} \overset{(2)}{K}_3 \overset{(2)}{K}_3 + 1584 \overset{(4)}{K}_{15} \overset{(2)}{K}_4 \overset{(2)}{K}_3 + 336 \overset{(4)}{K}_{15} \overset{(2)}{K}_3 \overset{(4)}{K}_4 \overset{(2)}{K}_3 + \\ & 12 \overset{(4)}{K}_3 \overset{(2)}{K}_4 \overset{(2)}{K}_3 + 480 \overset{(4)}{K}_{15} \overset{(2)}{K}_4 \overset{(2)}{K}_3 + 48 \overset{(4)}{K}_3 \overset{(2)}{K}_4 \overset{(2)}{K}_3 + 48 \overset{(4)}{K}_4 \overset{(2)}{K}_3 + 13824 \overset{(4)}{K}_{15} p^2 - \\ & 1152 \overset{(4)}{K}_{15} \overset{(2)}{K}_3 \overset{(2)}{p^2} - 128 \overset{(4)}{K}_{15} \overset{(2)}{K}_3 \overset{(2)}{p^2} + 13824 \overset{(4)}{K}_{15} \overset{(2)}{K}_4 \overset{(2)}{p^2} - 1152 \overset{(4)}{K}_{15} \overset{(2)}{K}_3 \overset{(4)}{K}_4 \overset{(2)}{p^2} - 768 \overset{(4)}{K}_{15} \overset{(2)}{K}_3 \overset{(4)}{K}_4 \overset{(2)}{p^2} - \\ & 32 \overset{(4)}{K}_3 \overset{(3)}{K}_4 \overset{(2)}{p^2} + 4896 \overset{(4)}{K}_{15} \overset{(2)}{K}_4 \overset{(2)}{p^2} - 672 \overset{(4)}{K}_{15} \overset{(4)}{K}_3 \overset{(4)}{K}_4 \overset{(2)}{p^2} - 120 \overset{(4)}{K}_3 \overset{(2)}{K}_4 \overset{(2)}{p^2} + 704 \overset{(4)}{K}_{15} \overset{(4)}{K}_3 \overset{(3)}{p^2} - \\ & 96 \overset{(4)}{K}_3 \overset{(4)}{K}_4 \overset{(3)}{p^2} + 32 \overset{(4)}{K}_4 \overset{(4)}{p^2} + 5184 \overset{(4)}{K}_{15} \overset{(3)}{K}_7 \overset{(2)}{p^2} - 864 \overset{(4)}{K}_{15} \overset{(2)}{K}_4 \overset{(4)}{K}_7 \overset{(2)}{p^2} - 288 \overset{(4)}{K}_{15} \overset{(2)}{K}_3 \overset{(4)}{K}_7 \overset{(2)}{p^2} + \\ & 3456 \overset{(4)}{K}_{15} \overset{(2)}{K}_4 \overset{(4)}{K}_7 \overset{(2)}{p^2} - 720 \overset{(4)}{K}_{15} \overset{(4)}{K}_3 \overset{(4)}{K}_4 \overset{(4)}{K}_7 \overset{(2)}{p^2} - 72 \overset{(4)}{K}_3 \overset{(2)}{K}_4 \overset{(4)}{K}_7 \overset{(2)}{p^2} + 684 \overset{(4)}{K}_{15} \overset{(2)}{K}_4 \overset{(4)}{K}_7 \overset{(2)}{p^2} - \\ & 126 \overset{(4)}{K}_3 \overset{(4)}{K}_4 \overset{(2)}{K}_7 \overset{(2)}{p^2} + 36 \overset{(4)}{K}_4 \overset{(3)}{K}_7 \overset{(2)}{p^2} + 432 \overset{(4)}{K}_{15} \overset{(2)}{K}_7 \overset{(2)}{p^2} - 144 \overset{(4)}{K}_{15} \overset{(4)}{K}_3 \overset{(4)}{K}_7 \overset{(2)}{p^2} + 144 \overset{(4)}{K}_{15} \overset{(4)}{K}_4 \overset{(4)}{K}_7 \overset{(2)}{p^2} - \\ & 36 \overset{(4)}{K}_3 \overset{(4)}{K}_4 \overset{(4)}{K}_7 \overset{(2)}{p^2} + 9 \overset{(4)}{K}_4 \overset{(2)}{K}_7 \overset{(2)}{p^2} + \sqrt{((4 \overset{(4)}{K}_{15} + \overset{(4)}{K}_4)^2 (38817792 \overset{(4)}{K}_{15}^6 p^4 + 3328 \overset{(4)}{K}_3^6 p^4 + \\ & 1920 \overset{(4)}{K}_3^5 p^2 (-\overset{(2)}{K}_3 + (13 \overset{(4)}{K}_4 + 6 \overset{(4)}{K}_7) p^2) + 1492992 \overset{(4)}{K}_{15}^5 (5 \overset{(2)}{K}_3 p^2 + 3 (13 \overset{(4)}{K}_4 + 5 \overset{(4)}{K}_7) p^4) + \\ & 48 \overset{(4)}{K}_3^4 (168 \overset{(2)}{K}_3^2 - 2 (155 \overset{(4)}{K}_4 + 36 \overset{(4)}{K}_7) \overset{(2)}{K}_3 p^2 + (1391 \overset{(4)}{K}_4^2 + 1320 \overset{(4)}{K}_4 \overset{(4)}{K}_7 + 264 \overset{(4)}{K}_7^2) p^4) + \\ & 62208 \overset{(4)}{K}_{15}^4 (33 \overset{(2)}{K}_3^2 + 2 (20 \overset{(4)}{K}_3 + 85 \overset{(4)}{K}_4 + 18 \overset{(4)}{K}_7) \overset{(2)}{K}_3 p^2 + (-104 \overset{(4)}{K}_3^2 + 559 \overset{(4)}{K}_4^2 + 420 \overset{(4)}{K}_4 \overset{(4)}{K}_7 + 66 \overset{(4)}{K}_7^2 - \\ & 4 \overset{(4)}{K}_3 (26 \overset{(4)}{K}_4 + 15 \overset{(4)}{K}_7) p^4) + \overset{(4)}{K}_4^2 (3328 \overset{(4)}{K}_4^4 p^4 + 81 \overset{(4)}{K}_7^4 p^4 + 216 \overset{(4)}{K}_4 \overset{(4)}{K}_7 \overset{(2)}{p^2} (4 \overset{(4)}{K}_3 + 3 \overset{(4)}{K}_7 p^2) + \\ & 1920 \overset{(4)}{K}_4^3 (4 \overset{(2)}{K}_3 p^2 + 3 \overset{(4)}{K}_7 p^4) + 144 \overset{(4)}{K}_4^2 (113 \overset{(2)}{K}_3^2 + 48 \overset{(4)}{K}_7 \overset{(2)}{K}_3 p^2 + 22 \overset{(4)}{K}_7^2 p^4) + \\ & 12 \overset{(4)}{K}_3^2 (416 \overset{(4)}{K}_4^4 p^4 + 108 \overset{(4)}{K}_7^4 p^4 + 54 \overset{(4)}{K}_4 \overset{(4)}{K}_7 \overset{(2)}{p^2} (-5 \overset{(2)}{K}_3 + 12 \overset{(4)}{K}_7 p^2) - 40 \overset{(4)}{K}_4^3 (104 \overset{(2)}{K}_3 p^2 - 57 \overset{(4)}{K}_7 p^4) + \\ & 9 \overset{(4)}{K}_4^2 (523 \overset{(2)}{K}_3^2 - 336 \overset{(4)}{K}_7 \overset{(2)}{K}_3 p^2 + 242 \overset{(4)}{K}_7^2 p^4) + 16 \overset{(4)}{K}_3^3 (4264 \overset{(4)}{K}_4^3 p^4 + 54 \overset{(4)}{K}_7^2 p^2 (-\overset{(2)}{K}_3 + 6 \overset{(4)}{K}_7 p^2) + \\ & \overset{(4)}{K}_4^2 (-2640 \overset{(2)}{K}_3 p^2 + 6525 \overset{(4)}{K}_7 p^4) + 9 \overset{(4)}{K}_4 (259 \overset{(2)}{K}_3^2 - 138 \overset{(4)}{K}_7 \overset{(2)}{K}_3 p^2 + 308 \overset{(4)}{K}_7^2 p^4) - \\ & 12 \overset{(4)}{K}_3 \overset{(4)}{K}_4 (1664 \overset{(4)}{K}_4^4 p^4 + 54 \overset{(4)}{K}_7^4 p^4 + 27 \overset{(4)}{K}_4 \overset{(4)}{K}_7 \overset{(2)}{p^2} (8 \overset{(2)}{K}_3 + 15 \overset{(4)}{K}_7 p^2) + \\ & 80 \overset{(4)}{K}_4^3 (16 \overset{(2)}{K}_3 p^2 + 39 \overset{(4)}{K}_7 p^4) + \overset{(4)}{K}_4^2 (-3036 \overset{(2)}{K}_3^2 + 1440 \overset{(4)}{K}_7 \overset{(2)}{K}_3 p^2 + 1848 \overset{(4)}{K}_7^2 p^4) - \\ & 3456 \overset{(4)}{K}_{15}^3 (208 \overset{(4)}{K}_3^3 p^4 - 3016 \overset{(4)}{K}_4^3 p^4 - 27 \overset{(4)}{K}_7^2 p^2 (\overset{(2)}{K}_3 + 3 \overset{(4)}{K}_7 p^2) - 15 \overset{(4)}{K}_4^2 (112 \overset{(2)}{K}_3 p^2 + 219 \overset{(4)}{K}_7 p^4) - \\ & 18 \overset{(4)}{K}_4 (35 \overset{(2)}{K}_3^2 + 39 \overset{(4)}{K}_7 \overset{(2)}{K}_3 p^2 + 55 \overset{(4)}{K}_7^2 p^4) + 12 \overset{(4)}{K}_3^2 (10 \overset{(2)}{K}_3 p^2 + (182 \overset{(4)}{K}_4 + 75 \overset{(4)}{K}_7) p^4) - \\ & 12 \overset{(4)}{K}_3 (6 \overset{(2)}{K}_3 + (50 \overset{(4)}{K}_4 + 9 \overset{(4)}{K}_7) \overset{(2)}{K}_3 p^2 - (169 \overset{(4)}{K}_4^2 + 165 \overset{(4)}{K}_4 \overset{(4)}{K}_7 + 33 \overset{(4)}{K}_7^2) p^4) + \\ & 432 \overset{(4)}{K}_{15}^2 (624 \overset{(4)}{K}_3^4 p^4 + 3744 \overset{(4)}{K}_4^4 p^4 + 27 \overset{(4)}{K}_7^4 p^4 + 162 \overset{(4)}{K}_4 \overset{(4)}{K}_7 \overset{(2)}{p^2} (\overset{(2)}{K}_3 + 2 \overset{(4)}{K}_7 p^2) - 80 \overset{(4)}{K}_3^3 \\ & (8 \overset{(2)}{K}_3 p^2 - 3 \overset{(4)}{K}_7 p^4) + 40 \overset{(4)}{K}_4^3 (88 \overset{(2)}{K}_3 p^2 + 129 \overset{(4)}{K}_7 p^4) + 9 \overset{(4)}{K}_4^2 (235 \overset{(2)}{K}_3^2 + 240 \overset{(4)}{K}_7 \overset{(2)}{K}_3 p^2 + 242 \overset{(4)}{K}_7^2 p^4) + \\ & 12 \overset{(4)}{K}_3^2 (39 \overset{(2)}{K}_3^2 - 4 (35 \overset{(4)}{K}_4 + 9 \overset{(4)}{K}_7) \overset{(2)}{K}_3 p^2 - 6 (91 \overset{(4)}{K}_4^2 + 70 \overset{(4)}{K}_4 \overset{(4)}{K}_7 + 11 \overset{($
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1	0	$ \begin{aligned} &(1728 \kappa_{15}^{(4)} \kappa_3^{(2)} + 576 \kappa_{15}^{(4)} \kappa_3^{(2)} K_3 + 48 \kappa_{15}^{(4)} \kappa_3^{(2)} K_3^2 + 48 \kappa_{15}^{(4)} \kappa_3^{(2)} K_3^3 + 1584 \kappa_{15}^{(4)} \kappa_3^{(2)} K_3^4 + 13824 \kappa_{15}^{(4)} p^2 - \\ &12 \kappa_3^{(4)} \kappa_4^{(2)} (K_3 + 480 \kappa_{15}^{(4)} \kappa_4^{(2)} K_3 + 48 \kappa_{15}^{(4)} \kappa_4^{(2)} K_3^2 + 48 \kappa_{15}^{(4)} \kappa_4^{(2)} K_3^3 + 13824 \kappa_{15}^{(4)} p^2 - \\ &1152 \kappa_{15}^{(4)} \kappa_3^{(2)} p^2 - 128 \kappa_{15}^{(4)} \kappa_3^{(2)} p^2 + 13824 \kappa_{15}^{(4)} \kappa_4 p^2 - 1152 \kappa_{15}^{(4)} \kappa_3 \kappa_4 p^2 - 768 \kappa_{15}^{(4)} \kappa_3 \kappa_4 p^2 - \\ &32 \kappa_3^{(4)} \kappa_4 p^2 + 4896 \kappa_{15}^{(4)} \kappa_4 p^2 - 672 \kappa_{15}^{(4)} \kappa_3 \kappa_4 p^2 - 120 \kappa_3^{(4)} \kappa_4 p^2 + 704 \kappa_{15}^{(4)} \kappa_4 p^2 - \\ &96 \kappa_3^{(4)} \kappa_4 p^2 + 32 \kappa_4^{(4)} p^2 + 5184 \kappa_{15}^{(4)} \kappa_7 p^2 - 864 \kappa_{15}^{(4)} \kappa_3 \kappa_7 p^2 - 288 \kappa_{15}^{(4)} \kappa_3 \kappa_7 p^2 + \\ &3456 \kappa_{15}^{(4)} \kappa_4 \kappa_7 p^2 - 720 \kappa_{15}^{(4)} \kappa_3 \kappa_4 \kappa_7 p^2 - 72 \kappa_3^{(4)} \kappa_4 \kappa_7 p^2 + 684 \kappa_{15}^{(4)} \kappa_4 \kappa_7 p^2 - \\ &126 \kappa_3^{(4)} \kappa_4 \kappa_7 p^2 + 36 \kappa_4^{(4)} \kappa_7 p^2 + 432 \kappa_{15}^{(4)} \kappa_7 p^2 - 144 \kappa_{15}^{(4)} \kappa_3 \kappa_7 p^2 + 144 \kappa_{15}^{(4)} \kappa_4 \kappa_7 p^2 - \\ &36 \kappa_3^{(4)} \kappa_4 \kappa_7 p^2 + 9 \kappa_4^{(4)} \kappa_7 p^2 - \sqrt{((\kappa_{15}^{(4)} + \kappa_4)^2 (38817792 \kappa_{15}^{(4)} p^4 + 3328 \kappa_3^{(4)} p^4 + \\ &1920 \kappa_3^{(4)} p^2 (-\kappa_3 + (13 \kappa_4 + 6 \kappa_7) p^2) + 1492992 \kappa_{15}^{(4)} (5 \kappa_3 p^2 + 3 (13 \kappa_4 + 5 \kappa_7) p^4) + \\ &48 \kappa_3^{(4)} (168 \kappa_3^{(2)} - 2 (155 \kappa_4 + 36 \kappa_7) \kappa_3 p^2 + (1391 \kappa_4^2 + 1320 \kappa_4 \kappa_7 + 264 \kappa_7^2) p^4) + \\ &62208 \kappa_{15}^{(4)} (33 \kappa_3^{(2)} + 2 (20 \kappa_3 + 85 \kappa_4 + 18 \kappa_7) \kappa_3 p^2 + (-104 \kappa_3^2 + 559 \kappa_4^2 + 420 \kappa_4 \kappa_7 + 66 \kappa_7^2) - \\ &4 \kappa_3 (26 \kappa_4 + 15 \kappa_7)) p^4) + \kappa_4^{(4)} (3328 \kappa_4^4 p^4 + 81 \kappa_7^4 p^4 + 216 \kappa_4 \kappa_7 p^2 (4 \kappa_3 + 3 \kappa_7 p^2) + \\ &1920 \kappa_4^3 (4 \kappa_3 p^2 + 3 \kappa_7 p^4) + 144 \kappa_4^2 (113 \kappa_3^{(2)} + 48 \kappa_7 \kappa_3 p^2 + 22 \kappa_7^2 p^4)) + \\ &12 \kappa_3^{(4)} (416 \kappa_4^4 p^4 + 108 \kappa_7^4 p^4 + 54 \kappa_4 \kappa_7 p^2 (-5 \kappa_3 + 12 \kappa_7 p^2) - 40 \kappa_4^3 (104 \kappa_3 p^2 - 57 \kappa_7 p^4) + \\ &9 \kappa_4^{(2)} (523 \kappa_3^{(2)} - 336 \kappa_7 \kappa_3 p^2 + 242 \kappa_7^2 p^4)) + 16 \kappa_3^{(4)} (4264 \kappa_4^3 p^4 + 54 \kappa_7^2 p^2 (-\kappa_3 + 6 \kappa_7 p^2) + \\ &\kappa_4^{(2)} (-2640 \kappa_3 p^2 + 6525 \kappa_7 p^4) + 9 \kappa_4 (259 \kappa_3^{(2)} - 138 \kappa_7 \kappa_3 p^2 + 308 \kappa_7^2 p^4)) - \\ &12 \kappa_3^{(4)} \kappa_4 (1664 \kappa_4^4 p^4 + 54 \kappa_7^4 p^4 + 27 \kappa_4 \kappa_7 p^2 (8 \kappa_3 + 15 \kappa_7 p^2) + \\ &80 \kappa_4^3 (16 \kappa_3 p^2 + 39 \kappa_7 p^4) + \kappa_4^2 (-3036 \kappa_3^{(2)} + 1440 \kappa_7 \kappa_3 p^2 + 1848 \kappa_7^2 p^4)) - \\ &3456 \kappa_{15}^{(4)} (208 \kappa_3^3 p^4 - 3016 \kappa_4^3 p^4 - 27 \kappa_7^2 p^2 (\kappa_3 + 3 \kappa_7 p^2) - 15 \kappa_4^2 (112 \kappa_3 p^2 + 219 \kappa_7 p^4) - \\ &18 \kappa_4 (35 \kappa_3^{(2)} + 39 \kappa_7 \kappa_3 p^2 + 55 \kappa_7^2 p^4) + 12 \kappa_3^{(4)} (10 \kappa_3 p^2 + (182 \kappa_4 + 75 \kappa_7) p^4) - \\ &12 \kappa_3^{(4)} (6 \kappa_3 + (50 \kappa_4 + 9 \kappa_7) \kappa_3 p^2 - (169 \kappa_4^2 + 165 \kappa_4 \kappa_7 + 33 \kappa_7^2) p^4)) + \\ &432 \kappa_{15}^{(4)} (624 \kappa_3^4 p^4 + 3744 \kappa_4^4 p^4 + 27 \kappa_7^4 p^4 + 162 \kappa_4 \kappa_7 p^2 (\kappa_3 + 2 \kappa_7 p^2) - 80 \kappa_3^{(4)} \\ &(8 \kappa_3 p^2 - 3 \kappa_7 p^4) + 40 \kappa_4^3 (88 \kappa_3 p^2 + 129 \kappa_7 p^4) + 9 \kappa_4^2 (235 \kappa_3^{(2)} + 240 \kappa_7 \kappa_3 p^2 + 242 \kappa_7^2 p^4) + \\ &12 \kappa_3^{(4)} (39 \kappa_3^2 - 4 (35 \kappa_4 + 9 \kappa_7) \kappa_3 p^2 - 6 (91 \kappa_4^2 + 70 \kappa_4 \kappa_7 + 11 \kappa_7^2) p^4) - 12 \kappa_3 \\ &(520 \kappa_4^3 p^4 + 27 \kappa_7^3 p^4 + \kappa_4^{(2)} (-80 \kappa_3 p^2 + 705 \kappa_7 p^4) - 3 \kappa_4 (25 \kappa_3^{(2)} + 6 \kappa_7 \kappa_3 p^2 - 88 \kappa_7^2 p^4))) + \\ &72 \kappa_{15}^{(4)} (832 \kappa_3^5 p^4 + \kappa_3^{(4)} (-560 \kappa_3 p^2 + 16 (247 \kappa_4 + 120 \kappa_7) p^4) + \\ &8 \kappa_3^{(4)} (147 \kappa_3^{(2)} - 10 (38 \kappa_4 + 9 \kappa_7) \kappa_3 p^2 + 2 (247 \kappa_4^2 + 285 \kappa_4 \kappa_7 + 66 \kappa_7^2) p^4) - \\ &2 \kappa_3^{(4)} (2912 \kappa_4^4 p^4 + 54 \kappa_7^4 p^4 + 54 \kappa_4 \kappa_7 p^2 (\kappa_3 + 9 \kappa_7 p^2) + 40 \kappa_4^3 (8 \kappa_3 p^2 + 123 \kappa_7 p^4) - \\ &9 \kappa_4^2 (163 \kappa_3^{(2)} - 48 \kappa_7 \kappa_3 p^2 - 286 \kappa_7^2 p^4)) - 4 \kappa_3^{(4)} (27 \kappa_7^2 \kappa_3 p^2 + 1144 \kappa_4^3 p^4 + 15 \kappa_4^2 (80 \kappa_3 p^2 + \\ &69 \kappa_7 p^4) - 198 \kappa_4 (4 \kappa_3 - 3 \kappa_7 \kappa_3 p^2 - \kappa_7^2 p^4)) + \kappa_4 (1664 \kappa_4^4 p^4 + 27 \kappa_7^4 p^4 + 27 \kappa_4 \kappa_7 p^2 \\ &(8 \kappa_3 + 9 \kappa_7 p^2) + 80 \kappa_4^3 (32 \kappa_3 p^2 + 33 \kappa_7 p^4) + 12 \kappa_4^2 (217 \kappa_3^{(2)} + 168 \kappa_7 \kappa_3 p^2 + 110 \kappa_7^2 p^4)))))))/ \\ &(((\kappa_{15}^{(4)} + \kappa_4) (12 \kappa_{15}^{(4)} - 4 \kappa_3 + \kappa_4) (6 \kappa_{15}^{(4)} + \kappa_3 + 2 \kappa_4)^2 \\ &\kappa_3) \end{aligned} $
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	1	$-\frac{6 \binom{2}{\kappa_3}}{-6 \binom{(4)}{\kappa_1+12} \binom{(4)}{\kappa_{15}-6} \binom{(4)}{\kappa_2-4} \binom{(4)}{\kappa_3+\kappa_4}}$	$-\frac{4}{6 \binom{(4)}{\kappa_1-12} \binom{(4)}{\kappa_{15}+6} \binom{(4)}{\kappa_2+4} \binom{(4)}{\kappa_3-\kappa_4}}$
	3	$-\frac{4 \binom{(4)}{\kappa_{15}+\kappa_3+2} \binom{(4)}{\kappa_4} \binom{(2)}{\kappa_3}}{4 \binom{(4)}{\kappa_2} \binom{(4)}{\kappa_{15}+\kappa_3+2} \binom{(4)}{\kappa_4} \binom{(4)}{\kappa_7}}$	$\frac{144 \binom{(4)}{\kappa_{15}}^2 + 4 \binom{(4)}{\kappa_3}^2 + 16 \binom{(4)}{\kappa_4}^2 + 8 \binom{(4)}{\kappa_4} \binom{(4)}{\kappa_7} + 3 \binom{(4)}{\kappa_7}^2 + 4 \binom{(4)}{\kappa_3} \binom{(4)}{\kappa_4 + \kappa_7} + 24 \binom{(4)}{\kappa_{15}} \binom{(4)}{2 \binom{(4)}{\kappa_3} + 4 \binom{(4)}{\kappa_4} + \binom{(4)}{\kappa_7}}}{\binom{(4)}{\kappa_{15} + \kappa_3 + 2} \binom{(4)}{\kappa_4} \binom{(4)}{2 \binom{(4)}{\kappa_3} + 2 \binom{(4)}{\kappa_4} - \binom{(4)}{\kappa_7}}^2}$
Decoupled unitarity condition(s)		(Does not apply to this case)	